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Pages: 2

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R,S) Exam April 2025 (2019 Scheme)

Course Code: EET468

Course Name: INDUSTRIAL INSTRUMENTATION AND AUTOMATION

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | Explain any three characteristics of transducers, | (3) |
| 2 | Explain the principle of a Resistance Potentiometer. | (3) |
| 3 | List the applications log amplifiers. | (3) |
| 4 | Write any 3 electrical control elements and its application. | (3) |
| 5 | List the advantages of virtual instrumentation systems. | (3) |
| 6 | List the key functions of process control network. | (3) |
| 7 | Discuss latching in PLC | (3) |
| 8 | Explain input elements and output elements in ladder logic. | (3) |
| 9 | Explain supervisory stations in SCADA and list its functions. | (3) |
| 10 | Describe a Remote Terminal Unit (RTU) and outline its functions. | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- | | | |
|----|---|-----|
| 11 | a) Sketch and explain the block diagram of process control loop in any industrial process. | (7) |
| | b) Explain first order and second order time responses of a sensor with relevant sketches and write the output equations. | (7) |

OR

- | | | |
|----|--|-----|
| 12 | a) Explain the criteria for the selection of a transducer for an industrial application. Mention five applications of transducers. | (7) |
| | b) Illustrate the principle and operation of variable reluctance tachometer. | (7) |

Module II

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- 13 a) Explain the operation of half wave and full wave precision rectifiers with relevant circuit diagrams. (8)
- b) Explain phase sensitive detectors and its significance. (6)

OR

- 14 a) Explain the operation of any two electrical actuators with schematic diagram. Mention their industrial applications. (8)
- b) Illustrate hydraulic actuating systems with its components used in industries. (6)

Module III

- 15 a) Compare radio and wireless communication. (6)
- b) With the help of a block diagram explain the architecture of virtual instrumentation. (8)

OR

- 16 a) Discuss about process control networks used in industries. (7)
- b) Explain the concepts and key features of graphical programming. (7)

Module IV

- 17 a) Sketch the block diagram of PLC and explain various components of PLC. (8)
- b) Explain about various counters used in PLC. (6)

OR

- 18 a) Sketch the ladder diagram for realising AND & OR logic using PLC. (8)
- b) Explain about various timers in PLC. (6)

Module V

- 19 a) Explain the architecture of SCADA in process automation. (8)
- b) Explain different control modes in DCS operation. (6)

OR

- 20 a) Illustrate the components of DCS architecture for industrial process monitoring and control. (8)
- b) Discuss the significance and features of DNP3 and IEC 60870-5-101 protocols used in SCADA systems. (6)
